Tossaporn (Tree) Saengja

tsaengja@gmail.com | tossatree.com
github.com/tj-14 | linkedin.com/in/tj14
(+66)62 406-0654

Eight years in medical AI — deep-learning research
(LLMs, medical imaging) and medical-device software.

education

Massachusetts Institute of Technology
Master of Engineering in Computer Science and Engineering 4.8/5.0 2019-2020
Bachelor of Science in Computer Science and Engineering 4.8/5.0 2015-2019
International Olympiad in Informatics 2013 (IOI), Bronze, 85th among 299 contestants (77 countries)
ACM Boston Preliminary 2016 (BOSPRE), 4^{th} place among 37 teams (16 schools)
IOI Thailand Coach (2023-Present), *Teaching Assistant for MIT 6.824: Distributed Systems (Prof. Robert Morris, Spring 2020)*

experience

SiData+, Siriraj Hospital — Data Scientist 2024-Present
- Medical reasoning Large Language Model (LLM) research with SCB 10x, focusing on improving general medical applications.
- Digital pathology: predicting BRCA1 germline mutation from H&E whole-slide images (MedSIGLIP), fine-tuned on a local cohort.
- Automated allergy patch-test grading from phone-camera photos (per-spot severity scoring) for the Dermatology department.
- Sarcopenia from CT: L3 selection (TotalSegmentator) and muscle segmentation (nnUNet) for Skeletal Muscle Index.

a2z Radiology — Software Engineer 2025-Present
- Built the radiologist DICOM viewer (Cornerstone3D), report editor, and prediction backend (DynamoDB).
- Built customer distribution (CLI, signed binaries, manifests) and the regulated device deployment pipeline on AWS Batch.
- Drove product security (pen-testing, vulnerability pipeline) and authored EU MDR / IEC 62304 submission and verification docs.

PreceptorAI, CARIVA — AI Consultant 2023-2025
- Developed machine learning models in collaboration with Siriraj Radiology Department:
  - Vision-Large Language Model to generate diagnostic reports from chest X-ray images (PyTorch).
  - Generative text-to-image model (diffusion model) for synthetic chest X-ray generation (MONAI).
- Led research and development of Gindee, a VLM based nutrition estimator.

Vision and Learning Lab, VISTEC — Research Assistant, under Prof. Supasorn Suwajanakorn 2023-2024
- Investigated object collocation by developing Score Distillation Sampling (SDS) to leverage Stable Diffusion's priors.
- Enhanced local attention masking in CLIP for zero-guidance segmentation.
- Conducted experiments on diffusion model memorization (CIFAR-10).

Mahidol Wittayanusorn High School (national top-tier science school) — Computer Science Teacher 2020-2023
- Coached students to represent Thailand in IOI 2022.
- Lectured data science (Python), web dev, blockchain, Computer Olympiad (C++), programming (Python).
- Advised year-long student projects, primarily in machine learning.

Agnos — Technical Cofounder 2019-2023
- Co-developed an AI-powered diagnosis system (Python, Django) used by physicians. (130k+ records, 50k+ active users)
- Led Data Science, AI, and Growth teams, helped securing 30M THB funding (Shark Tank Thailand S3).

Facebook, Notification Backend — Intern 2017
- Implemented a pipeline for network effect prediction (classification/regression) using HiveQL and FBLearner Flow.
- Increased comment metrics by 0.5% in production (2B active users).

MIT Media Lab, Human Dynamics — Undergrad Researcher, under Yan Leng 2018-2020
- Developed an optimization framework (Python) for learning network structure and marginal benefits from 400GB+ datasets.

publications

On the Robustness of Answer Formats in Medical Reasoning Models *(co-corresponding)*
P. Taveekitworachai*, N. Pongjirapat*, K. Chaisutyakorn, P. Ittichaiwong, **T. Saengja****, K. Pipatanakul**. arXiv:2509.20866
LUSD: Localized Update Score Distillation for Text-Guided Image Editing *(co-first)*
W. Chinchuthakun*, **T. Saengja***, N. Tritrong, P. Rewatbowornwong, P. Khungurn, S. Suwajanakorn. ICCV 2025
ZeroGuideSeg++: Improving Local Attention Masking in CLIP for Zero-Guidance Segmentation
P. Rewatbowornwong, N. Chatthee, **T. Saengja**, E. Chuangsuwanich, S. Suwajanakorn.
Decomposing Food Images for Better Nutrition Analysis
P. Khlaisamniang, et al. (**T. Saengja** as corresponding author). CVPR MetaFood 2025
GPU Poor's Guide to Radiology Report Generation
K. Udomlapsakul, et al. (**T. Saengja** as third author). Proc. BioNLP 2024.

certifications

CITI Program — Good Clinical Practice & Human Subjects Research, Basic Stage. 2025–2029

reviewers

NeurIPS 2026, ECCV 2026, ICLR 2026, CVPR 2026